

Lecture 33

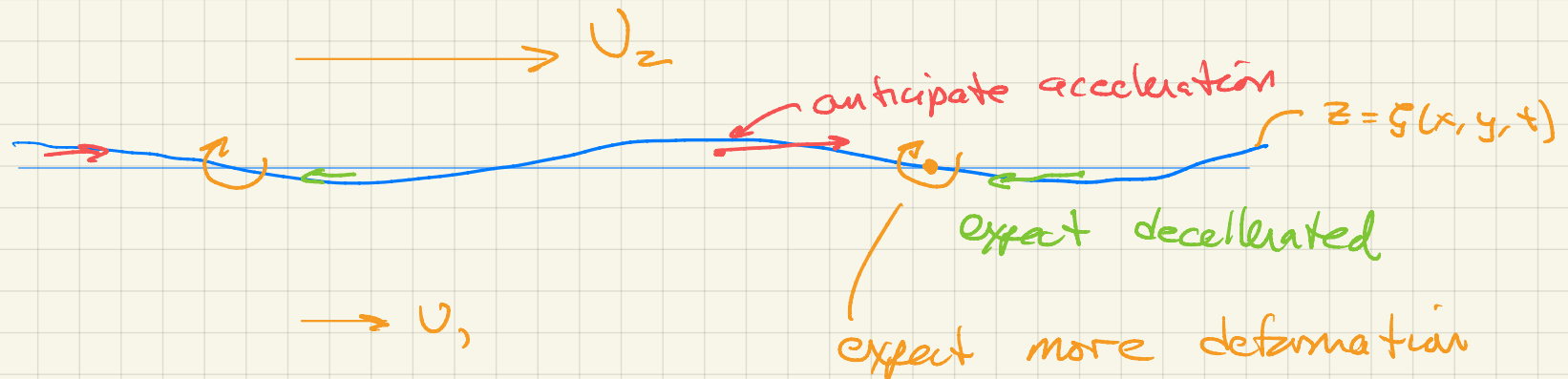
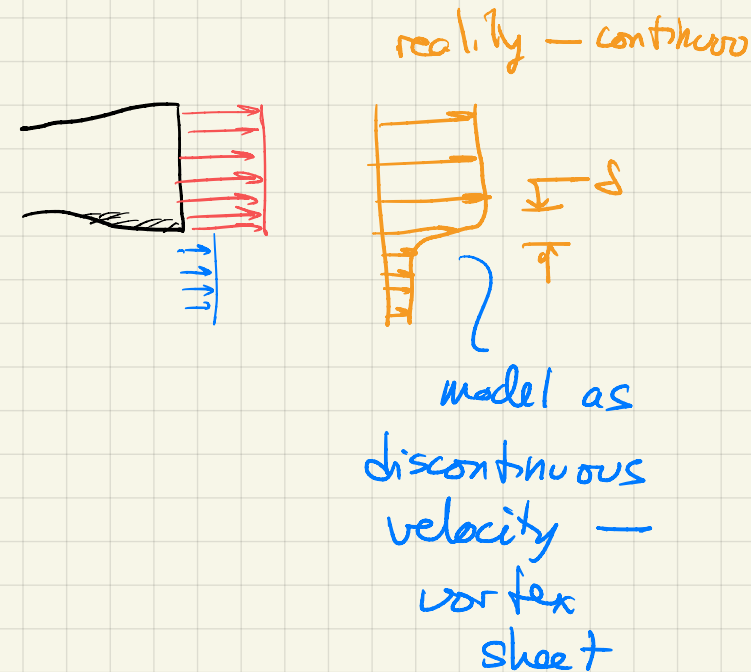
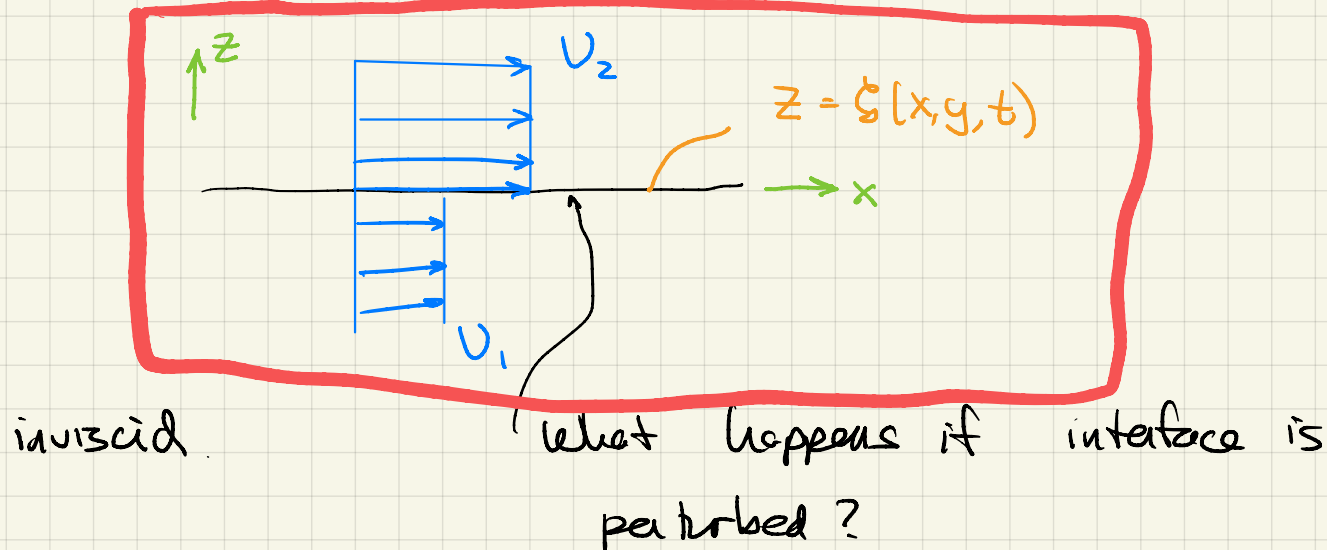
Stability

- Kelvin - Helmholtz
- Dispersion relations
- Stability questions...
spatial, temporal

- indifference Reynolds # and experiments
 ↳ below which perturbations decay

- transient growth finite time instability of
 nominally $t \rightarrow \infty$ stable flows

Kelvin-Helmholtz Instability



Linearized

$$\phi_1 = \overbrace{U_1 x}^{\text{base flow}} + \phi_1'$$

base flow $\frac{\partial U_1 x}{\partial x} = U_1$

$$\phi_2 = U_2 x + \underbrace{\phi_2'}_{\text{perturbation, small, } O(\epsilon)}$$

$\rightarrow \epsilon$ small, $O(\epsilon)$

$$\underbrace{f(z=\epsilon)}_{\text{applying BC @ } z=\epsilon} = \underbrace{f(z=0)}_{\text{applying at } z=0} + \underbrace{\epsilon \frac{\partial f}{\partial z} \bigg|_{z=0}}_{\text{perturbation, small, } O(\epsilon)} + O(\epsilon)^2$$

applying
BC @
 $z=\epsilon$



applying at
 $z=0$

To leading order
can apply BC
at $z=0$

• Linearize gov eq:

$$\nabla^2 \phi_1 = 0$$

$$\nabla^2 (U_1 x + \phi_1') = 0$$

$$\begin{aligned} \nabla^2 \phi_1' &= 0 \\ \nabla^2 \phi_2' &= 0 \end{aligned}$$

gov. eq.

• Kinematic BC ($z=0$)

$$\frac{\partial \phi}{\partial z} = \frac{\partial \phi}{\partial t} + \frac{\partial \phi}{\partial x} \frac{\partial \zeta}{\partial x} + \frac{\partial \phi}{\partial y} \frac{\partial \zeta}{\partial y} \quad (\text{from above})$$

$$\frac{\partial (\cancel{u_x} + \phi')}{\partial t} = \frac{\partial (\cancel{u_x} + \phi')}{\partial t} + \frac{\partial (\cancel{u_x} + \phi')}{\partial x} \frac{\partial \zeta}{\partial x} + \frac{\partial (\cancel{u_x} + \phi')}{\partial y} \frac{\partial \zeta}{\partial y}$$

$O(\epsilon^2)$
 $O(\epsilon^2)$

$$\frac{\partial \phi'}{\partial z} = \frac{\partial \phi'}{\partial t} + U \frac{\partial \zeta}{\partial x} + O(\epsilon^2)$$

$$\begin{aligned} \frac{\partial \phi'_1}{\partial z} &= \frac{\partial \phi'_1}{\partial t} + U_1 \frac{\partial \zeta}{\partial x} \\ \frac{\partial \phi'_2}{\partial z} &= \frac{\partial \phi'_2}{\partial t} + U_2 \frac{\partial \zeta}{\partial x} \end{aligned}$$

Kinematic

• dynamic BC

$$\frac{\partial \phi_1'}{\partial t} = O(\epsilon)$$

$$C_1 - \frac{1}{2} |\nabla \phi_1|^2 - \frac{\partial \phi_1}{\partial t} = C_2 - \frac{1}{2} |\nabla \phi_2|^2 - \frac{\partial \phi_2}{\partial t}$$

$$\uparrow$$

$$|\nabla (U_1 + \phi_1')|^2$$

$$(U_1 + \frac{\partial \phi_1'}{\partial x})^2 + \left(\frac{\partial \phi_1'}{\partial y}\right)^2 + \left(\frac{\partial \phi_1'}{\partial z}\right)^2$$

$$U_1^2 + 2U_1 \frac{\partial \phi_1'}{\partial x} + \left(\frac{\partial \phi_1'}{\partial x}\right)^2$$

$\sim 1 \quad \sim \epsilon \quad \sim \epsilon^2$

$$\nabla \phi_1' \rightarrow 0 \text{ at } z \rightarrow \infty$$

$$\nabla \phi_2' \rightarrow 0 \text{ at } z \rightarrow \infty$$

→ $O(1)$:

$$C_1 - \frac{1}{2} U_1^2 = C_2 - \frac{1}{2} U_2^2$$

✓ P balance
±∞

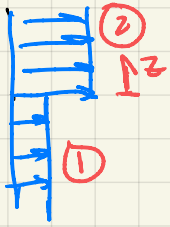
→ $O(\epsilon)$

$$U_1 \frac{\partial \phi_1'}{\partial x} + \frac{\partial \phi_1'}{\partial t} = U_2 \frac{\partial \phi_2'}{\partial x} + \frac{\partial \phi_2'}{\partial t}$$

dyn.
b.c.

NEW ↓

assume general form of perturbations



$$\begin{bmatrix} \phi_1'(x, y, z, t) \\ \phi_2'(x, y, z, t) \\ \zeta(x, y, t) \end{bmatrix} = \begin{bmatrix} \hat{\phi}_1(z) \\ \hat{\phi}_2(z) \\ \hat{\zeta} \end{bmatrix} e^{ikx + i\ell y + st}$$

wave character

Fourier if $k, \ell \in \text{Real}$
exp growth / decay if $s \in \text{Real}$

plug in: $\nabla^2 \phi' = 0$

$$\frac{\partial^2 \phi_1'}{\partial x^2} + \frac{\partial^2 \phi_1'}{\partial y^2} + \frac{\partial^2 \phi_1'}{\partial z^2} = 0$$

$$\underbrace{(-k^2 - \ell^2)}_{-\tilde{k}^2} \hat{\phi}_1 + \frac{d^2 \hat{\phi}_1}{dz^2} = 0$$

$$\frac{d^2 \hat{\phi}_1}{dz^2} - \tilde{k}^2 \hat{\phi}_1 = 0$$

$$\hat{\phi}_1(z) = A_1^+ e^{+\tilde{k}z} + A_1^- e^{-\tilde{k}z}$$

for decay $z \rightarrow -\infty \Rightarrow A_1^- = 0$

$$\hat{\phi}_1(x) = A_1^+ e^{+\tilde{k}z} \quad \dots \quad \hat{\phi}_2(x) = A_2^- e^{-\tilde{k}z}$$

$$e^{ikx - i\omega t}$$

propagating wave

$\parallel$

$$e^{ik(x - \frac{\omega}{k}t)}$$

points of constant phase —
peak of a wave

$$x = \frac{\omega}{k}t + (\text{phase const})$$



$$c_{ph} = \frac{\omega}{k}$$

phase speed

Still need to figure out

- ① how A_1^+, A_2^- relate to $\hat{\xi}$?
- ② which k, l, s "work"?

apply BC

from above

$$\frac{\partial \phi_1'}{\partial z} = \frac{\partial \xi}{\partial t} + U_1 \frac{\partial \xi}{\partial x}$$

$$\frac{\partial \phi_2'}{\partial z} = \frac{\partial \xi}{\partial t} + U_2 \frac{\partial \xi}{\partial x}$$

Sub in again ...

$$A_1 \tilde{k} = s \hat{\xi} + U_1 i k \hat{\xi} \\ (\text{all } x \text{ } e^{ikx + i ly + st})$$

$$-A_2 \tilde{k} = s \hat{\xi} + U_2 i k \hat{\xi} \\ (\text{all } x \text{ } e^{ikx + i ly + st})$$

$$A_1 = \frac{s + ik U_1}{\tilde{k}} \hat{\xi}$$

$$A_2 = - \frac{s + ik U_2}{\tilde{k}} \hat{\xi}$$

$$U_1 \frac{\partial \phi_1'}{\partial x} + \frac{\partial \phi_1'}{\partial t} = U_2 \frac{\partial \phi_2'}{\partial x} + \frac{\partial \phi_2'}{\partial t}$$

$$A_1 (ik U_1 + s) = A_2 (ik U_2 + s)$$

$$(\text{all } x \text{ } \hat{\xi} e^{ikx + i ly + st})$$

all $z \rightarrow 0$

$$e^{\tilde{k} z} = 1, e^{-\tilde{k} z} = 1$$

$$\left(\frac{s + ikU_1}{R} \right) (ikU_1 + s) = - \left(\frac{s + ikU_2}{R} \right) (ikU_2 + s)$$

an s, k relationship

solve for s in terms of k

$$s^2 + 2ikU_1 s - k^2 U_1^2 = -(s^2 + 2ikU_2 s - k^2 U_2^2)$$

$$2s^2 + 2ik(U_1 + U_2)s - k^2 U_1^2 - k^2 U_2^2 = 0$$

↑
- ?

solve quadratic (fix sign??)

$$s = \underbrace{ik \frac{(U_1 + U_2)}{2}}_{S_i} \pm \underbrace{\frac{k}{2} \sqrt{(U_1 - U_2)^2}}_{S_r}$$

$\times e^{st}$

if $k \in \mathbb{R}$

↑

wave behaviour

$$e^{iks + is\tau}$$

↓

↑

growth / decay

$$\times (e^{s-\tau})$$

$$\cos(kx + \omega t)$$

$$C_{ph} = -\frac{\omega}{k}$$

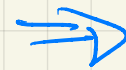
speed of
points
of
const
phas

$$x = -\frac{\omega}{k} t + (\text{phase})$$

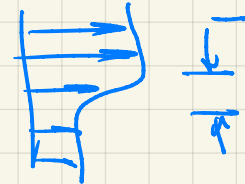
for S_r (growth/decay) $\rightarrow$ faster growth for larger k

something
else happens
when wave length
 $\approx$ thickness

$$S_r \sim k \quad (\text{short wave length})$$



in reality,



more general case ... $\rho_1 \neq \rho_2$

stabilizing

$$\frac{\omega}{k} = -i \frac{\rho_1 U_1 + \rho_2 U_2}{\rho_1 + \rho_2} \pm \left[\underbrace{-\frac{g}{k} \frac{\rho_1 - \rho_2}{\rho_1 + \rho_2}}_{\text{from putting "gravity" in gov. eq.}} + \frac{\rho_1 \rho_2}{(\rho_1 + \rho_2)^2} (U_1 - U_2)^2 \right]^{1/2}$$

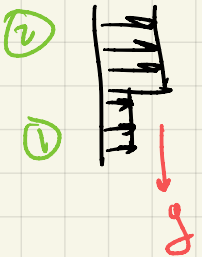
negative enough

$\rightarrow$ negative

argument of Γ

$\rightarrow$ pure imaginary S

$\rightarrow$ pure wave
behavior



What we did is fairly general for stability analysis....

subbed in $e^{ikx + i\ell y + st}$

got $D(s, k, \ell) = 0$ ~~✗~~ dispersion relation...

Above we asked about "temporal stability"

took $k \in \text{Reals}$ $\rightarrow$ just wavelength $\frac{2\pi}{\lambda} = k$

$\rightarrow$ periodic in x

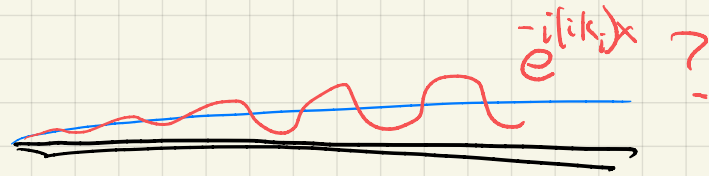
$\rightarrow e^{ikx}$

solved for s

$\rightarrow$ real part: amplification

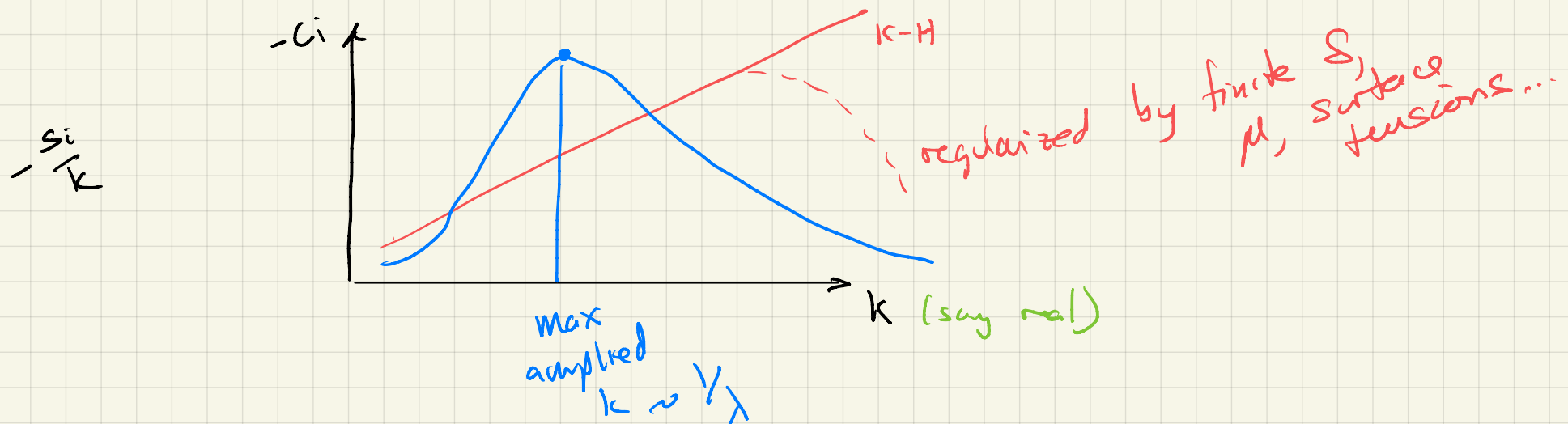
$\rightarrow$ imaginary part: $\text{Cph} = -\frac{s_i}{k}$
(related to wave/phase speed)

We could have asked about spatial instability
 — how disturbances amplify in space



- take $s = i\omega$ ω real (a frequency)
 - solve for complex $k = k_r + ik_i$
- amplification down stream

way to look at results



600

include additional parameters ...

Re ?

$$\Rightarrow D(k, c, Re) = 0$$

Reynolds # dependent
dispersion relation

Q: at what Re is
any wave length
unstable

